MILITARY SPECIFICATION

RELAY PANEL, FIRING

(For Torpedo Mark 37 Mods 0, 2, and 3)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification describes the requirements for the procurement, inspection, and testing of Firing Relay Panel, LD 629583.

20 APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-621	Box; Wood, Nailed and Lock Corner

Military

MIL-P-116	Preservation, Methods of
MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

FSC 1355

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MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base

DRAWINGS

Naval Ordnance Systems Command

LD 629583	Firing Relay Panel Assembled, and All Parts and Subassemblies Listed Thereon
Drawing 2445177	Schematic Diagram

PUBLICATIONS

Naval Ordnance Systems Command

OS 6341	Ordnance Design Requirements; General, Miscellaneous

(Copies of specifications, standards, drawings, and Publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw-Thread Standards for Federal. Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Description. The firing relay panel consists of a double-pole, double-throw relay and a double-pole, single-throw control circuit relay. The firing relay operates from 115 volts ac, 60 Hz input and the control' circuit switching relay operates from 26 volts dc. These are the switching relays necessary for a torpedo run. The firing relay is closed by the pulse

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that fires the primary battery squibs in the warshot condition, and the control relay, when energized, connects the battery to the control circuitry.

3.2 General requirements

3.2.1 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproduction and -periodic production testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2.2 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the firing relay panel shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing firing relay panels that comply with the technical requirements of the contract. See 4.3.1.

3.2.3 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the firing relay panel shall conform to the detailed requirements indicated on the drawings and specifications listed in LD 629583.

3.2.4 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions for the firing relay panel shall be in accordance with the applicable drawings, specifications, and publications. listed in LD 629583.

3.2.5 Moisture, fungus, and corrosion. The firing relay panel shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified in the drawings and specifications listed in LD 629583. When moisture, fungus, and corrosion resistance are not specified in these documents, the item shall conform to OS 6341.

3.2.6 Marking. Marking of assemblies subassemblies, and parts shall be in accordance with the applicable drawings listed in LD 629583.

3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

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3.3 Performance requirements and product characteristics. The firing relay panel shall meet the following performance requirements and product characteristics:

3.3.1 The firing relay panel shall operate on the following relay coil voltages:

3.3.1.1 115 volts plus or minus 5 percent, 60 plus or minus 2 Hz, single phase, ac.

3.3.1.2 26 plus or minus 3 volts dc.

3.3.2 All relay pin, and Connector designations referred to herein are given on Drawing 2 45177.

3.3.3 With 115 volts ac (as specified in 3.3.1.1) applied between pins A and B of connector P721, continuity shall exist between pins C and K of P721. An open circuit shall exist between pins E and I of P721.

3.3.4 With 26 volts dc (as specified in 3.3.1.2) applied between pins F and H of P721, continuity shall exist between pins K and L of P721, and between pins N and P of P721.

3.3.5 With no applied voltages across the terminals energized in 3.3.3 and 3.3.4, the continuities between the respective pins given therein shall. be open circuits. A closed circuit shall exist between pins E and I of P721.

3.3.6 There shall be at least 50 megohms resistance between all pins of P721 and the frame of the firing relay Panel. (Not applicable after temperature and humidity testing.) After temperature and humidity testing, the insulation resistance shall be at least 25 megohms from pins A, B, C, E, and I to the frame of the firing relay panel and at least one megohm from pins D, F, E, K, L, N, and P of P721 to the frame of the firing relay panel.

3,4 Environmental requirements. The firing relay panel shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.4.

3.4.1 Vibration. The firing relay panel shall meet the following vibration requirements:

3.4.1.1 Vibration, nonoperating, The firing relay panel shall show no damage and-shall meet all the performance requirements and product

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characteristics of this specification after being tested as in 4.4.1.2.

3.4.1.2 Vibration, operating. The firing relay panel shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.4.1.3.

3.4.1.3 Vibration, transportation. The firing relay panel shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.1.4.

3.4.2 Shock, transportation. The firing relay panel shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.2.

3.4.3 Temperature and humidity

3.4.3.1 High temperature and humidity nonoperating. The firing relay panel shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.3.1.

3.4.3.2 High temperature, operating. The firing relay panel shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.3.2.

3.4.3.3 Low temperature, nonoperating. The firing relay panel shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.3.3.

3.4.3.4 Low temperature, operating. The firing relay panel shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.3.4.

3.5 Cleanup. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

3.6 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, impregnation of coils, plating, painting, machine screw assemblage, welding, riveting, brazing, and freedom of parts from all burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performante of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the firing relay panel shall be classified as follows:

(a) Preproduction inspection (paragraph 4.4)

(b) Quality conformance inspection (paragraph 4.5)

(c) Periodic production inspection (paragraph 4.4)

4.3 sampling

4.3.1 Preproduction. A preproduction sample of five firing relay panels, manufactured in accordance with 3.2.2, shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects In the sample. Further production of the relay panels by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.5, a random sample of five firing relay panels will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The firing relay panels shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

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4.4 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.5, shall be subjected to the following tests and examinations, Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.4.1 Vibration

4.4.1.1 Mounting. The firing relay panel, less pack and packaging, shall be mounted rigidly, secured with the mounting base horizontal to a vibration device, using the four mounting holes in the base and tested as specified in 4.4.1.2 and 4.4.1.3.

4.4.1.2 Vibration, nonoperating. The firing relay panel shall be subjected to the "Vibration, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.1.1.

4.4.1.3 Vibration, operating. The firing relay panel shall be subjected to the "Vibration, operating" test of MIL-E-82590 and during and thereafter shall meet the requirements of 3.4.1.2.

4.4.1.4 Vibration, transportation. The firing relay panel shall be subjected to the "Vibration, transportation" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.1.3.

4.4.2 Shock, transportation. The firing relay panel shall be subjected to the "Shock, transportation" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.2.

4.4.3 Temperature and humidity

4.4.3.l High temperature and humidity, nonoperating. The firing relay panel shall be subjected to the "High temperature and humidity, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.3.1.

4.4.3.2 High temperature, operating. The firing relay panel shall be subjected to the "High temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.4.3.2.

4.4.3.3 Low temperature, nonoperating. The firing relay panel shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.3.3.

4.4.3.4 Low temperature, operating. The firing relay panel shall be subjected to the "Low temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.4.3.4.

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4.4.4 Waiver of tests. Tests specified in 4.4.1.4 and 4.4.2 above may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.4.5 Functional tests. At the conclusion of the tests specified in 4.4.1.4 and 4.4.2 above, functional and other tests, as applicable, shall be conducted on the firing relay panel to determine freedom from operational malfunction and compliance with all performance requirements and product characteristics of this specification.

4.5 Quality conformance inspection. Each firing relay panel delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.5.1 Test conditions. Unless otherwise specified, the firing relay panels shall be subjected to quality conformance tests under the following conditions:

4.5.1.1 Temperature: Room ambient 65 degrees F to 95 degrees F (18.33 degrees C to 35 degrees C).

4.5.1.2 Altitude: Normal ground.

4.5.1.3 Vibration: Normal ground.

4.5.1.4 Humidity: From 0 to 95 percent relative maximum, inclusive.

4.5.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required tests.

4.5.3 Test equipment. The following items of test equipment are required to perform the quality conformance tests set forth in this specification:

4.5.3.1 DC power supply, 20 volts minimum, 22 volts maximum, at five amperes.

4.5.3.2 AC power supply, 110 volts minimum, 113 volts maximum, 60 plus or minus 2 Hz, single phase, at 2 amperes.

4.5.3.3 Megohmmeter, 500 volts, capable of indicating resistances of 50 megohms.

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4.5.3.4 Continuity meter.

4.5.4 Test procedure

4.5.4.1 All pin designations specified herein are on connector P721, as shown on Drawing 2445177.

4.5.4.2 Make continuity checks as given in Table 1, Continuity shall be as indicated there in.

TABLE I

From Pin	To Pin	Continuity
D	P	Closed circuit
D	N	Open circuit
c	L	Open circuit
F	H	Relay coil resistance
c	K	Open circuit
K	L	Open circuit
N	P	Open circuit
E	I	Closed circuit

4.5.4.3 Apply 110 VOltS ac (see 4.5.3.2) across pins A and B and repeat the measurements in Table I. The readings shall be unchanged except for pin C to pin K, which shall be a closed circuit, and pin E to pin I, which shall be an open circuit.

4.5.4.4 Disconnect the ac power supply from pins A and13, connect 21 volts dc (4.5.3.1) across pins F and H, and make the continuity measurements given in Table II. Continuity shall be as indicated therein.

TABIE II

From Pin	To Pin	Continuity
D	P	Closed circuit
D	N	Closed circuit
C	L	Open circuit
C	K	Open circuit
K	L	Closed circuit
N	P	Closed circuit
E	I	Closed circuit

4.5.4.5 Using a 500 volt xnegohmmeter (4.5.3.3), measure from each pin of PT21 to a point on the firing relay panel frame. The firing relay panel shall meet the requirements specified in 3.3.6.

4.5,4.6 Visual and dimensional examination. Each firing relay panel shall be examined to determine conformance with the requirements of workmanship and dimensions.

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4.5.4.7 Packaging, packing, and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be *as* specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. The firing relay panel shall be packaged in accordance with Method IIB of MIL-P-116.

5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Unit packaging. The firing relay panel shall be packaged in accordance with Method III of MIL-P-116.

5.2 packing. Packing shall be Level A or Level C as specified (see 6.2).

5.2.1 Level A

5.2.1.1 Consolidation container. The firing relay panel, packaged as described in 5.1.1.1, shall be packed in a wooden box in accordance with PPP-B-621, Class 2.

5.2.1.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Pattern No. 101 (MS 53102). (See 6.2.)

5.2.2 Level C. The firing relay panel, packaged in accordance with 5.1.3.1, shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Levels A or C cushioning. The unit package shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to ensure the firing relay panel will meet all the requirements of Section 3 after being tested in accordance with paragraph 4.4.1.4 (Vibration), and 4.4.2 (Shock). At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent.

5.4 Marking

5.4.1 Special markings. None, unless otherwise specified.

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5.4.2 Normal markings. In addition to the markings required by the contract or order, the unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use. The firing relay Panel is intended for use in Torpedoes Mk 37 Mods 0, 2, and 3.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions *to* this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned activity for preproduction and periodic production inspections.

(f) Data required other than that specified in 3.2.1.

(g) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

6.2.1 The attention of the-contracting officer is invited to the quality assurance provisions and options in MIL-P-116 and to 4.4.4.

Preparing Activity
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